MOONBASE

4-DAY SURVIVAL MISSION

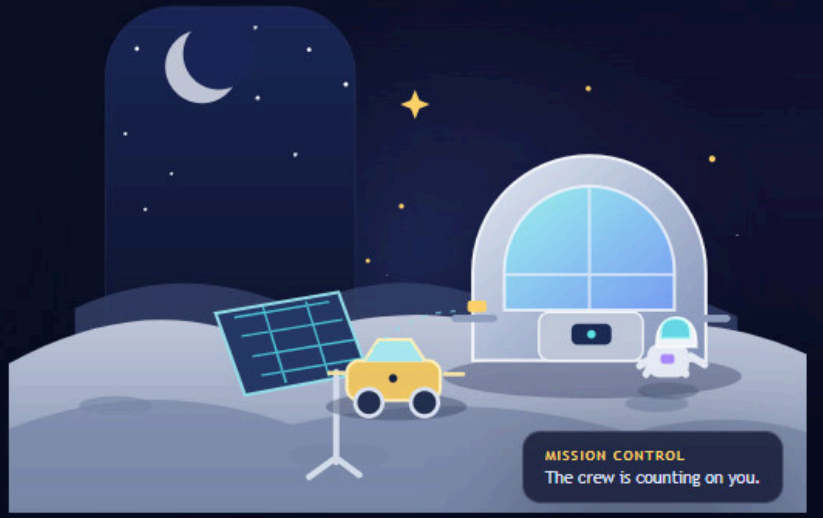
A MISSION FOR YOUNG EXPLORERS

Your MoonBase. Four days to survive.

The crew needs your help. Make smart choices, protect the base, and learn how real space teams plan for life beyond Earth.

Four mission days • science-inspired choices • no saved progress

Start mission →



01 Keep four base supplies in balance

02 Choices can help now or later

03 Every mission begins fresh

An educational project for the NASA Space Apps Challenge

Made for curious explorers ages 8-12

BEFORE LAUNCH

Mission briefing

Guide the crew through four mission days. Each round uses game-value supplies, and your earlier repairs can change what happens next.

Your objective

Keep Power, Water, and Food above zero, then finish Day 4 with a ready shelter. Daily needs are used before each decision. Choose a plan that helps today and leaves the crew ready for tomorrow.

How to play

Read the situation and science note. Select one choice, review its trade-off, and confirm it. Mission Control calculates every supply change.

Power · 12 / 12
Runs the habitat, rover, water recycler, and grow lights.

Water · 12 / 12
Keeps the crew supplied; recycling can stretch the reserve.

Food · 10 / 12
The crew uses food each mission day; a delayed grow-light harvest can help.

Shelter readiness · 8 / 12
An abstract readiness meter for the storm shelter, not a radiation dose.

This is an educational game, not a real engineering simulator or an official NASA product.
Refreshing the page starts you at the beginning.

Begin Day 1

MISSION DAY 1 OF 4

Base status

Power	Water	Food	Shelter readiness
12 / 12	12 / 12	10 / 12	8 / 12
Runs the habitat, rover, water recycler, and grow lights.	Keeps the crew supplied; recycling can stretch the reserve.	The crew uses food each mission day; a delayed grow-light harvest can help.	An abstract readiness meter for the storm shelter, not a radiation dose.
Game value · game units	Game value · game units	Game value · game units	Game value · game units

DAY 1 EVENT

Dust on the solar array

The rover spots dust on the solar array, and mission control asks the crew to prepare for a possible solar-weather alert. Use the work window to boost power later or prepare the shelter.

SCIENCE SIGNAL

Lunar dust can cling to exposed surfaces and make solar-array power harder to maintain. NASA has an Orion plan to use stored supplies as a shelter during a solar-particle event.

Source: Lunar Dust Considerations for Vertical Solar Arrays Volume 1: Lunar Environment and Dust Interactions

Source: Dust Mitigation for Flexible Solar Arrays (DMFlex-ACO)

Source: How NASA Will Protect Astronauts From Space Radiation at the Moon

Choose the crew's plan

SELECTED

Brush the solar array

Spend 1 Power now to clear the panels.

The array adds 1 Power on each of Days 2-4.

SELECT THIS PLAN

Connect the spare battery

Gain 1 Power now without spending rover power.

You get energy right away, but there is no later array boost.

SELECT THIS PLAN

Prepare the storm shelter

Spend 1 Power and gain 2 Shelter readiness.

The prepared shelter avoids normal readiness wear on Days 2-4.

Ready to send your choice to Mission Control.

Confirm decision →

Mission effects

No delayed effects yet. Your choices can change future days.

MISSION RULE

Daily needs are applied before your choice. Supplies use a 0-12 game-value scale, and the backend checks each update.

✓

DAY 1 DECISION REPORT

Brush the solar array

The rover used power to clear the array. Its improved output is scheduled for the next three mission days.

Every resource is a game value. A careful plan keeps the base ready for the next challenge.

Power	Water	Food	Shelter readiness
9 / 12	10 / 12	8 / 12	7 / 12
-3 this day	-2 this day	-2 this day	-1 this day
Before 12 · daily -2 · delayed 0 · event 0 · choice -1	Before 12 · daily -2 · delayed 0 · event 0 · choice 0	Before 10 · daily -2 · delayed 0 · event 0 · choice 0	Before 8 · daily -1 · delayed 0 · event 0 · choice 0

Days 2-4: the cleaned array adds 1 Power each day.

Continue to Day 2

MISSION DAY 2 OF 4

Base status

Power	Water	Food	Shelter readiness
9 / 12	10 / 12	8 / 12	7 / 12
Runs the habitat, rover, water recycler, and grow lights.	Keeps the crew supplied; recycling can stretch the reserve.	The crew uses food each mission day; a delayed grow-light harvest can help.	An abstract readiness meter for the storm shelter, not a radiation dose.
Game value · game units	Game value · game units	Game value · game units	Game value · game units

DAY 2 EVENT

The water loop needs care

The water recovery filter is due for service. You can spend power to improve the loop or open a sealed backup pack.

SCIENCE SIGNAL

NASA's International Space Station water system filters and treats recovered water. It demonstrated a 98% recovery goal; this game's Water points do not use that percentage.

Source: NASA Achieves Water Recovery Milestone on International Space Station

Choose the crew's plan

SELECTED

Service the recycler

Spend 1 Power and gain 1 Water now.

Water use is 1 lower on Days 3 and 4.

SELECT THIS PLAN

Open a backup water pack

Spend 3 Power and gain 5 Water now, up to the storage limit.

It uses no Power, but it does not reduce later water use.

Ready to send your choice to Mission Control.

Confirm decision →

Mission effects

- Solar array cleaned: +1 Power on Days 2-4.

MISSION RULE

Daily needs are applied before your choice. Supplies use a 0-12 game-value scale, and the backend checks each update.

✓

DAY 2 DECISION REPORT

Service the recycler

The crew spent a little power and gained water by servicing the recycler. The improved loop reduces water use on Days 3 and 4.

Every resource is a game value. A careful plan keeps the base ready for the next challenge.

Power	Water	Food	Shelter readiness
7 / 12	9 / 12	6 / 12	6 / 12
-2 this day	-1 this day	-2 this day	-1 this day
Before 9 · daily -2 · delayed +1 · event 0 · choice -1	Before 10 · daily -2 · delayed 0 · event 0 · choice +1	Before 8 · daily -2 · delayed 0 · event 0 · choice 0	Before 7 · daily -1 · delayed 0 · event 0 · choice 0

Days 3-4: water use drops by 1 game unit each day.

Continue to Day 3

MISSION DAY 3 OF 4

Base status

Power	Water	Food	Shelter readiness
7 / 12	9 / 12	6 / 12	6 / 12
Runs the habitat, rover, water recycler, and grow lights.	Keeps the crew supplied; recycling can stretch the reserve.	The crew uses food each mission day; a delayed grow-light harvest can help.	An abstract readiness meter for the storm shelter, not a radiation dose.
Game value · game units	Game value · game units	Game value · game units	Game value · game units

DAY 3 EVENT

Fresh food for the crew

The crew is choosing how to support the food supply. An LED grow tray can help later, while a meal crate helps right away.

SCIENCE SIGNAL

NASA's Advanced Plant Habitat uses LED lights and carefully delivered water and nutrients to study growing plants in space. A harvest takes much longer in real life than one game turn.

Source: Advanced Plant Habitat

Choose the crew's plan

SELECTED

Start the LED grow tray

Spend 2 Power and 1 Water to start the tray.

Gain 3 Food on Day 4 as a compressed game-time effect.

SELECT THIS PLAN

Open the meal crate

Gain 3 Food now from stored meals.

This helps right away and creates no later harvest.

Ready to send your choice to Mission Control.

Confirm decision →

Mission effects

- Solar array cleaned: +1 Power on Days 2-4.
- Recycler serviced: Water use is 1 less on Days 3-4.

MISSION RULE

Daily needs are applied before your choice. Supplies use a 0-12 game-value scale, and the backend checks each update.

✓

DAY 3 DECISION REPORT

Start the LED grow tray

The grow tray used power and water today. Its small harvest is scheduled for the next mission day.

Every resource is a game value. A careful plan keeps the base ready for the next challenge.

Power	Water	Food	Shelter readiness
4 / 12	7 / 12	4 / 12	5 / 12
-3 this day	-2 this day	-2 this day	-1 this day
Before 7 · daily -2 · delayed +1 · event 0 · choice -2	Before 9 · daily -2 · delayed +1 · event 0 · choice -1	Before 6 · daily -2 · delayed 0 · event 0 · choice 0	Before 6 · daily -1 · delayed 0 · event 0 · choice 0

Day 4: the grow tray adds 3 Food game units.

Continue to Day 4

✓

DAY 4 DECISION REPORT

Power the full shelter setup

The crew powered the full shelter setup and added stored supplies as shielding.

Every resource is a game value. A careful plan keeps the base ready for the next challenge.

Power	Water	Food	Shelter readiness
2 / 12	6 / 12	5 / 12	6 / 12
-2 this day	-1 this day	+1 this day	+1 this day
Before 4 · daily -2 · delayed +1 · event 0 · choice -1	Before 7 · daily -2 · delayed +1 · event 0 · choice 0	Before 4 · daily -2 · delayed +3 · event 0 · choice 0	Before 5 · daily -1 · delayed 0 · event -2 · choice +4

No extra effect is scheduled for a later day.

See mission results

MISSION STATUS · COMPLETE

Four days. One brave crew. Mission complete!

The crew finished all four mission days with supplies and a ready storm shelter.

Power 2 / 12

Runs the habitat, rover, water recycler, and grow lights.

Game value · game units

Water 6 / 12

Keeps the crew supplied; recycling can stretch the reserve.

Game value · game units

Food 5 / 12

The crew uses food each mission day; a delayed grow-light harvest can help.

Game value · game units

Shelter readiness 6 / 12

An abstract readiness meter for the storm shelter, not a radiation dose.

Game value · game units

Final base supplies

Power · 2 / 12
Game-value units

Water · 6 / 12
Game-value units

Food · 5 / 12
Game-value units

Shelter readiness · 6 / 12
Game-value units

What the crew learned

- Power, water, food, and safety planning work together.
- Repairing a system can cost resources now and save them later.
- NASA's Orion plan uses stored supplies to make a temporary shelter during a solar-particle event. This game's lunar shelter value is only a game measure.

Mission choices

DAY 1	Brush the solar array — Dust can reduce solar-array performance; maintenance can repay its energy cost later.
DAY 2	Service the recycler — A repaired recovery system can save water on later days.
DAY 3	Start the LED grow tray — A powered grow tray uses water and energy before its later food benefit.
DAY 4	Power the full shelter setup — A stronger shelter plan uses more power and gives more readiness.

Start a new mission

Back to the welcome screen